PITCH ONLY (30,000 FT)

```
% States: alpha (rad), V (m/s), q (rad/s), theta (rad)
% Inputs: elevator (deg)

A = [-0.588001538590337,-
0.000485130200553057,0.963643379459107,0.000373646126261995; ...
     6.46507102085155,-0.0114437025203809,-0.109376182414917,-
9.80971536694793; ...
     -4.06783934380150,-1.35331065699645e-07,-0.861234994633703,0; ...
     0,0,1,0];

B = [-0.00113780424173252;0.00666450903970976;-0.125819481907869;0];

C = eye(4);
D = zeros(4,1);
```

Obtained by linearising the 4th-order nonlinear model at the following flight condition:

```
alpha = 0.0969406483519791; % rad
theta = alpha; % rad
V = 200; % m/s
q = 0; % rad/s

de = -6.56892175518988; % deg
CG = 25; % %MAC
T  = 10692.6988729931; % N

rho = 0.458312441644953 % air density in kg/m3
```

ROLL ONLY (SEA LEVEL)

```
% States: beta (rad), p (rad/s), r (rad/s), phi (rad)
% Inputs: aileron (deg), elevator (deg), rudder (deg)

A = [[0.0302582801646847,0.0385920582400911,-
0.991187901030574,0.0490139813877847]; ...
             [-50.4394400206613,-4.25791953695668,0.413517871751583,0]; ...
             [12.7950410566372,-0.0247684391372182,-0.738832397029264,0];
...
             [0,1.0,0.0383440783846401,4.02611936928295e-10]];

B = [[0,0,0.00108553786818762];...
             [-1.19823787862564,0,0.234586481629371];...
             [-0.0347276122866948,0,-0.117750756847346];...
             [0,0,0]];

C = eye(4);
D = zeros(4,3);
```

Obtained by linearising the 8th-order nonlinear model at the following flight condition:

```
alpha = 0.038325304315471; % rad
theta = alpha; % rad
V = 200; % m/s
q = 0; % rad/s
```

```
da = 0; % deg
de = -4.90116289413029; % deg
dr = 0; % deg
CG = 25; % %MAC
T  = 28160.8767522768; % N

rho = 1.225 % air density in kg/m3
```

And keeping only the 4 states of interest: beta, p, r, and phi.